Case F

Mr. Walters — Chronic Non-Specific Low Back Pain | Working Dx: CNSLBP with Central Sensitisation & Fear-Avoidance Behaviour

Target: 70%+ | Musculoskeletal (MSK) Physiotherapy

■ PROBLEM LIST

- Generalised lumbar stiffness across all planes
- Poor transversus abdominis and multifidus activation
- Reduced cardiovascular fitness and physical deconditioning
- Obesity (BMI 34) increasing mechanical spinal load
- Central sensitisation — amplified, disproportionate pain response
- Likely hip flexor tightness from prolonged driving career
- Ceased all walking / dog walking activity
- Avoids all lifting — functionally disabling
- Reduced social participation and daily activity
- Loss of retirement role and identity
- Low self-efficacy for pain management and physical activity

■ RED FLAGS

- Cauda Equina Syndrome (Bowel / Bladder Dysfunction, Saddle Anaesthesia)
- Unexplained Weight Loss / Night Pain / Systemic Symptoms (Malignancy or Infection)
- History of Osteoporosis / Significant Trauma (Vertebral Fracture Risk)
- Progressive Neurological Deficit (New Leg Weakness, Sensory Loss)

■ PATHOPHYSIOLOGY

- In this state, the dorsal horn neurones of the spinal cord become sensitised such that non-noxious stimuli (normal movement, light touch, routine posture changes) are amplified and perceived as painful.
- Walters experiences constant pain despite no structural findings on imaging.
- The result is a pain system that is "turned up" independently of tissue state.
- Fear-avoidance behaviour operates as a powerful amplifier of this central sensitisation.
- Walters' belief that his spine is crumbling activates the limbic system, increasing amygdala-mediated threat appraisal.

■ DIFFERENTIAL DIAGNOSES

1. Supporting features:

- Chronic Non-Specific LBP with Central Sensitisation & Fear-Avoidance
- Supporting features:

2. Supporting features:

- Lumbar Spondylosis / Facet Joint Osteoarthritis
- Supporting features:
- Age 65, 30-year truck driving career (compressive load), dull aching LBP, generalised lumbar stiffness. Degenerative changes are extremely common at this age and occupational history.

3. Supporting features:

- Lumbar Canal Stenosis (Neurogenic Claudication)
- Supporting features:
- Age 65, male, bilateral leg discomfort worsened by walking and prolonged standing, relieved by sitting or flexion (shopping trolley sign). Prior occupation and obesity increase degenerative risk.

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Tampa Scale of Kinesiophobia (TSK-17)

What it measures: A validated 17-item self-report questionnaire measuring pain-related fear of movement and (re)injury. Scores range from 17 to 68 — higher scores indicate greate | How to perform: 1. Provide the patient with the Tampa Scale questionnaire — it can be completed in

Active Lumbar Range of Motion with Pain Behaviour Observation

What it measures: Quantifies lumbar mobility in all planes; observes movement quality, pain behaviour, and inconsistency between active and passive ROM — a potential sign of fear | How to perform: 1. Patient stands. Ask to flex forward ("reach towards your toes") — measure finger

Repeated Movement Assessment (McKenzie Method — Directional Preference)

What it measures: Whether repeated movements in a specific direction (flexion or extension) centralise, peripheralise, or have no effect on symptoms — identifies if a mechanical | How to perform: 1. Perform 10 repeated end-range lumbar extensions in standing (or prone press-ups i

■ TREATMENT PLAN

■ Initial Phase — Weeks 1–4: Pain Education, Graded Exposure & Core Reactivation

- "Mr. Walters, your scan being normal is actually very good news — it means there is nothing broken or crumbling. The pain is real, but it's coming from an overactive alarm system in your nervous system, not from damage." — Directly addresses the catastrophising belief. Evidence shows PNE reduces kinesiophobia, catastrophising, and healthcare utilisation in CNSLBP. The "alarm system" metaphor is accessible and non-threatening.

• "Pain is not always a sign of damage. In your case, your nervous system has become sensitised after years of guarding your back — it has been doing its job a little too well." — Introduces central sensitisation in lay terms. This begins to decouple the pain-equals-damage association that drives kinesiophobia.

• "The best thing we can do for your back is to gradually start moving more. Movement is actually medicine for the nervous system — it helps calm it down and rebuild strength." — Directly counters the avoidance behaviour. Frames exercise as a treatment, not a risk.

• "We will go at your pace. We will set goals together, and I will explain everything we do and why — you are in control of this process." — Restores Mr. Walters' sense of agency and internal locus of control. This is therapeutically essential in fear-avoidance: empowerment over passivity.

• Tampa Scale score reduced by ≥5 points (meaningful reduction in kinesiophobia)

■ Long-Term Plan — Weeks 4–12: Functional Strengthening, Lifestyle & Self-Management

• Progressive core strengthening programme: Advance from TrA activation → dead bug exercises → bird-dog → modified plank → functional lifting mechanics (hip hinge with load). Goal is to restore confidence with loaded tasks, not just muscle strength in isolation. Frame each progression as evidence to Mr. Walters that his spine is strong and safe.

• Graded lifting exposure: Systematic hierarchical exposure to lifting tasks using a fear ladder (identify feared activities, order from least to most feared, work through each with physiotherapist support). The specific goal — returning to grocery lifting and routine household tasks — should be named explicitly as treatment targets on the PSFS.

• Obesity management: Refer to dietitian for personalised dietary advice. Encourage swimming or hydrotherapy as a low-joint-load cardiovascular option given BMI 34. Even 5–10% weight reduction meaningfully reduces lumbar compressive load and systemic inflammation. Frame this positively, not as criticism.

• Interdisciplinary input: If TSK remains high at 4 weeks despite education, refer to a clinical psychologist or pain management service for Acceptance and Commitment Therapy (ACT) or Cognitive Behavioural Therapy (CBT) — both are evidence-based for CNSLBP with fear-avoidance. Coordinate with GP regarding sleep management and mood screening.

• Self-management independence: Transition from therapist-led to fully independent home programme. Provide written, illustrated exercise plan. Use a walking log or app to foster accountability. Consider peer support groups for chronic pain (e.g. community pain education programmes).